

From Next-Word Prediction to Interlanguage Infilling: Adapting GLMs for Learner Cloze Simulation

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Abstract

Artificial learners — language models adapted to second-language (L2) writing — are increasingly used as backbones for learner simulation, proficiency assessment, and exercise generation. The dominant backbone is a decoder-only next-word-prediction (NWP) model. We argue that this choice is wrong for the tasks teachers and assessment systems actually use to probe learner knowledge: *fill-the-gap* and other cloze items. NWP cannot condition on the right context of the gap; it is biased toward fluent target-language completions rather than learner-plausible ones; and its cloze accuracy collapses when the surrounding context itself contains learner errors. We propose instead that learner cloze simulation should be grounded in an *autoregressive blank-infilling* objective, and adapt the GLM framework of Du et al. (2022) to learner data. With a lightweight learner-conditioning prefix (L1, CEFR, ERRANT error-profile bucket), a continued-pretrained GLM becomes an *Interlanguage Language Model* (ILM): a model whose predictive distribution over a cloze gap approximates the empirical filler distribution a real learner at a specified proficiency would produce. On a held-out EFCAMDAT cloze set, the ILM matches or outperforms a strong NWP baseline on exact match while reducing KL divergence to the empirical learner-filler distribution by a large margin, with the gap widening under noisy learner context and at multi-token gaps — the regime the paper is about. We further evaluate transfer to CELVA-SP, KUPA-KEYS, and andrew100k, and find that distribution-level metrics degrade less than mode-level metrics, consistent with the paper’s central claim that *density*, not *mode*, is the right object for learner cloze simulation.

1 Introduction

Cloze and fill-the-gap exercises are the most common format in which second-language (L2) assessment and practice actually reach learners. A realistic artificial-learner system — one that could

be used to simulate student responses, calibrate item difficulty, or generate plausible distractors — must therefore model learner responses to cloze items well. The de-facto backbone used for artificial learners in recent work is a decoder-only *next-word-prediction* (NWP) model such as GPT-2 (Radford et al., 2019) or a larger instruction-tuned causal LM, sometimes with supervised fine-tuning on learner corpora (Caines et al., 2023; Naismith et al., 2023).

We argue this default is wrong, and for structural reasons that compound:

1. **NWP cannot see the right context of a gap.** Cloze items are infilling tasks: the correct filler depends jointly on left and right context. NWP factorises the distribution left-to-right; it can either discard the right context (variant 04) or smuggle it in via prompt engineering (variant 03), but both are workarounds for an objective mismatch.
2. **NWP is biased toward the *native* filler.** Trained on large, overwhelmingly native corpora, a decoder-only LM concentrates probability mass on the modal target-language completion. A learner at B1 does not produce the modal completion; they produce a filler drawn from a *different* distribution — one that NWP does not directly model.
3. **NWP collapses in learner context.** When the text surrounding the gap contains the learner’s own upstream/downstream errors — the natural setting for any real cloze constructed from authentic learner writing — NWP’s already-weak distributional estimate degrades further.

The *autoregressive blank infilling* objective proposed in Du et al. (2022) removes the first pathology by construction: its Part A attends bidirectionally, its Part B generates the masked spans autoregressively, and the two are unified in a single

Transformer. GLM was introduced as an NLU-vs-generation unification; we repurpose it as the right *objective* for learner cloze simulation. 133

Pathologies (2) and (3) are not solved by GLM alone: a GLM pretrained on Wikipedia still produces native-filler distributions. We therefore adapt GLM to learner data in two steps: *continued pre-training* on EFCAMDAT with the GLM objective, and a lightweight *learner-conditioning prefix* that exposes L1, CEFR level, and a coarse ERRANT-profile bucket to the model. We call the resulting object an **Interlanguage Language Model (ILM)** — a language model whose predictive distribution over a cloze gap approximates the distribution a real L2 learner with the specified profile would produce. 146

Contributions 147

1. A **diagnostic evaluation** showing that NWP-based infillers — in both left-to-right and prompt-based-fill-the-blank configurations — fail as learner cloze models in precisely the regime where cloze items are interesting: non-initial gaps, multi-token spans, and authentic learner context (§6). 154
2. A **method**, *ILM*, that adapts the GLM objective to learner text via continued pretraining and a learner-conditioning prefix, and that matches the NWP baseline on exact match while drastically improving *learner-plausibility* and *filler-distribution calibration* (§4, §7). 161
3. An **SLA-construct evaluation** at L2 error loci (determiners, prepositions, verb morphology, subject-verb agreement) showing that ILM’s per-locus learner-plausibility pattern matches the predictions of Goldschneider and DeKeyser (2001) and Pienemann (1998); NWP baselines produce a flat surface (§8). 168
4. **Cross-corpus transfer** experiments on CELV-SP, KUPA-KEYS, and andrew100k, which show that distribution-level metrics (KL, learner-plausibility) transfer more reliably than model-level metrics (exact match), consistent with the claim that density, not mode, is the right object (§9). 174

2 Related Work 176

Artificial learners and learner simulation. A line of work trains or adapts language models on

L2 corpora to produce “artificial learners”: models that generate learner-like errors (Caines et al., 2023; Naismith et al., 2023; Chollampatt and Ng, 2018). Recent analyses establish construct validity for such models by showing that their error distributions align with SLA-predicted typologies (Anonymous, 2026a, our CMCL-2026 companion work). This paper extends the artificial-learner programme from free-generation error distributions to the cloze / infilling setting. 094

Cloze and fill-the-gap in NLP for education. 095

Cloze items are widely used for L2 assessment and practice (Taylor, 1953). Prior automated approaches rely on MLMs like BERT (Devlin et al., 2019) or on NWP prompted to “fill the blank” (Panda et al., 2022; Kurdi et al., 2020). We show that both approaches inherit objective mismatches for the learner case. 100

Infilling objectives. T5 (Raffel et al., 2020) introduced span-denoising seq-to-seq pretraining; BART (Lewis et al., 2020) used denoising with encoder-decoder. Donahue et al. (2020) and Shen et al. (2020) study text infilling directly. GLM (Du et al., 2022) unifies autoencoding and autoregressive generation through autoregressive blank infilling with 2D positional encodings and span shuffling. We adopt GLM as our backbone precisely because of its autoregressive, variable-length, bidirectional-context span generation — the combination required for learner cloze. 113

Metadata- and persona-conditioned LMs. 113

Conditioning LMs on learner metadata for generation and assessment has been explored via prompt-based control and prefix-tuning (Anonymous, 2026b, our sibling sketch). We transplant that conditioning framing from NWP to the GLM objective. 120

3 Background: GLM 121

Du et al. (2022) propose the autoregressive blank-infilling objective. Given input $x = [x_1, \dots, x_n]$, multiple spans $\{s_1, \dots, s_m\}$ are sampled (Poisson span lengths, $\lambda=3$, 15% of tokens masked). Each span is replaced by a single [MASK] token in Part A. Part B consists of the masked spans in a shuffled order, each prefixed with [START] and suffixed with [END]. The model is trained to maximise 126

$$\max_{\theta} \mathbb{E}_{\mathbf{z} \sim Z_m} \left[\sum_{i=1}^m \log p_{\theta}(s_{z_i} \mid x_{\text{corrupt}}, \mathbf{s}_{\mathbf{z}_{<i}}) \right],$$

with Part A attending bidirectionally and Part B attending to Part A plus its own antecedents. This single objective subsumes MLM (single-span, length 1), prefix-LM (single-span, length > 50%), and T5-style span-denoising (multi-span, independent positional ids), without the architectural asymmetry of encoder-decoder models. The 2D positional encoding removes the requirement that the model know the span length in advance, which is the pathology that afflicts BERT on multi-token blanks

For our purposes, two properties matter:

1. **Bidirectionality over the corrupted text.** The filler is conditioned on both left and right of the gap. This is what a cloze item requires
2. **Autoregressive span generation.** Multi-token fillers are generated with full intra-span dependency, unlike MLM’s independence assumption.

4 Method: From GLM to ILM

4.1 Objective

We retain GLM’s autoregressive blank-infilling objective verbatim, but apply it to learner corpora with a conditioning prefix.

4.2 Learner-conditioning prefix

Every training instance and every inference-time cloze item is prepended in Part A with:

[L1=lang] [CEFR=level] [ERRPROF=k]

where:

- L1 is the writer’s first language (EFCAMDAT metadata; UNK when unavailable);
- CEFR is the writer’s proficiency label (A1...C2);
- ERRPROF is a coarse bucket (16 clusters) means over the ERRANT error-type vector (on EFCAMDAT) imputed at inference for corpora without explicit metadata.

4.3 Continued pretraining

Starting from a public GLM checkpoint (GLM-Base 110M; GLM-RoBERTa-Large 335M), we continue-pretrain on EFCAMDAT with the GLM objective (Poisson $\lambda=3$, 15% mask, span shuffle) and the conditioning prefix. Three epochs, effective batch size 256; lr 10^{-5} ; AdamW; linear warm-up 5%.

4.4 Cloze inference

For an evaluation item with left context L , gap g , right context R , and metadata $m = (L1, CEFR, ERRPROF)$, we build Part A as

[L1=...] [CEFR=...] [ERRPROF=...] L [MASK] R

and greedily (or via nucleus sampling) generate the contents of the first span in Part B. The model’s predictive distribution over the gap is the object we evaluate.

5 Experimental Setup

5.1 Main experiment (variant 01)

The main experiment instantiates the following cell of the design space:

Backbone	GLM + learner-conditioning prefix
Gap type	contiguous multi-token spans ($\lambda=3$)
Context	authentic learner context (condition II)
Training regime	continued pretraining on EFCAMDAT

Remaining variants are ablations: vanilla GLM without conditioning, NWP prompt baseline, NWP L→R baseline, BERT MLM baseline, L2-error-loci evaluation, span-length ablation, context-condition ablation, training-regime ablation, and cross-corpus transfer. See the supplementary experiments-ideas/ for full per-variant specifications.

5.2 Evaluation set

We evaluate on a **100-item CEFR-stratified sample** of EFCAMDAT-test (20 items per level, A1 through C1), with one contiguous multi-token gap per item of length 1–3 (Poisson $\lambda=2$, max length 3). All items are in condition II (authentic learner context surrounding the gap). The sample is explicit and reproducible: it is materialised as `samples/efcamdat-smoke-100.csv` by a seeded sampler and is checked into the codebase alongside the derived cloze JSONL (`cloze.jsonl`, same content across systems).

5.3 Data

Primary. EFCAMDAT, CEFR-stratified train/test split following the splits under `/phd-experimental-data/cefr-classification/data/splits/`.

Transfer. CELVA-SP, KUPA-KEYS, andrew100k, universal-CEFR, under the same splits directory. Each is turned into a multi-token cloze

test set by sampling spans of length 1–6 at positions drawn uniformly over the text (so gaps are not biased to sentence-initial).

Native-filler reference and clean-context counterfactuals. We generate a GEC-corrected version of every text with CoEdit-Large (Rabeja et al., 2023), which defines condition (I) “clean” and supplies the native reference filler for each gap. Condition (II) is the raw learner text; condition (III) applies calibrated synthetic corruption to condition (I) to match the ERRANT error rate of EFCAMDAT-B1.

Not a priority. C4-200M is mentioned only as a potential scaling control; it is not run for this paper.

5.4 Baselines

- **NWP-L→R** (variant 04): decoder-only LM, left context only, greedy continuation.
- **NWP-prompt** (variant 03): decoder-only LM with a fill-the-blank prompt template; zero-shot and SFT on EFCAMDAT cloze triples
- **MLM** (variant 05): BERT/DeBERTa with k mask tokens (length known) and iterative unmasking (length unknown).
- **GLM-vanilla** (variant 02): GLM continued-pretrained on EFCAMDAT without the conditioning prefix.

5.5 Metrics

- **Exact match / Top- k** against the gold learner filler.
- **Learner-plausibility** $LP = \log p_{\theta}(f_{\text{learner}} | x_{\text{corrupt}}) - \log p_{\theta}(f_{\text{native}} | x_{\text{corrupt}})$ — positive LP means the model prefers the learner filler.
- **Filler-distribution calibration.** For items where multiple learner fillers exist at matched CEFR, we report $D_{\text{KL}}(p_{\text{empirical}} || p_{\theta})$ and JS divergence.
- **Robustness gap.** $\Delta = \text{metric(II)} - \text{metric(I)}$.
- **Cross-corpus degradation.** Per-target $\delta = \text{metric(target)} - \text{metric(EFCAMDAT-test)}$.

6 Diagnostic: Where NWP and Vanilla MLM Fail

6.1 Baseline = model $\times$ decoding strategy

An encoder-only MLM, a decoder-only NWP, and a GLM do not admit the same notion of “run cloze”. The MLM must be wrapped in some decoding strategy to produce more than one subtoken; the NWP

can generate greedily until EOS; GLM has autoregressive span generation built into its objective. We therefore treat a **baseline as a (model, decoding-strategy) pair** and report both axes explicitly. The strategies we exercise on the 100-item smoke are listed in Table 1.

6.2 Overall EM by (model $\times$ decoding)

Finding 1 — NWP-prompt collapses to the template. distilgpt2 with nwp_greedy_prompt achieves EM **1.0%**: inspection shows it *echoes the underscore* from the prompt (e.g. gold “Hi! This is” $\rightarrow$ predicted “_____ the menu:”). Without SFT on the fill-the-blank format, a decoder-only LM does not treat “_____” as a gap.

Finding 2 — MLM single-subtoken predictions dominate in absolute EM but hide a structural failure. MLM-none outperforms every NWP baseline on overall EM (roberta-base 28.0%, bert-base-cased 24.0%). This is superficially surprising until it is stratified by gold subtoken length (Table 3): MLM-none achieves **0% EM on every gap whose gold text tokenises to > 1 subtoken in the model’s own tokenizer**. The overall advantage is entirely carried by single-subtoken gaps (e.g. articles, prepositions) where “return one subtoken” is sufficient by *construction*.

Finding 3 — A decoding strategy is a system component, not a detail. Pairing the same distilbert-base-uncased with mlm_iterative_confident (17.0%) instead of mlm_none (16.0%) is a different system; the gain comes from the former reconstructing multi-subtoken spans at all. The paper treats every reported baseline as a (model, decoding) pair; conflating the two is exactly the failure mode mlm_none surfaces.

6.3 Per-CEFR and per-length stratification

Table 4 shows EM stratified by gap length in NLTK TreebankWordTokenizer units (the default reference tokenizer in §5.3). The multi-token collapse is universal: every non-GLM baseline hits 0 EM at $L \geq 3$.

These findings define the failure mode ILM is designed to fix: NWP cannot exploit the right context of the gap (Finding 2 strata), and its prompt-based workaround needs SFT even to produce well-formed outputs (Finding 1). Encoder-only MLMs under mlm_none are architecturally incapable of predicting multi-subtoken spans at all. The

Strategy	Family	Description
mlm_none	MLM	Single [MASK] regardless of gap length; top-1 subtoken. Emits exactly one subtoken.
mlm_iterative_confident	MLM	Length-known: k masks where k = gap length in model subtokens; fill most-confident first (Mask-Predict).
nwp_greedy_prompt	NWP	Fill-the-blank prompt, greedy decoding.
nwp_greedy_l2r	NWP	Left context only, greedy continuation.
glm_ar_span	GLM	Autoregressive Part B generation until [END].

Table 1: Decoding strategies. The strategy is part of the experimental unit: reporting “an MLM achieves X” without naming the strategy conflates (model) with (model×decoding), which hides the main pathology documented below.

Model	Decoding	EM (%)
distilgpt2 (82M)	prompt	1.0
gpt2 (124M)	l2r	14.0
distilbert-base-unc. (66M)	mlm_none	16.0
bert-base-uncased (110M)	mlm_none	20.0
bert-base-cased (110M)	mlm_none	24.0
roberta-base (125M)	mlm_none	28.0
distilbert-base-unc. (66M)	iter-conf	17.0
glm-roberta-large (335M)	ar-span	17.0

Table 2: Zero-shot EM on the 100-item EFCAMDAT-test smoke; no fine-tuning, no conditioning prefix. Each row is a distinct (model, decoding) unit. Short names: prompt = nwp_greedy_prompt; l2r = nwp_greedy_l2r; iter-conf = mlm_iterative_confident; ar-span = glm_ar_span (Table 1). GLM uses its native build_inputs_for_generation + eop_token_id protocol for infilling; the reported number is not yet conditioned on learner metadata.

MLM backbone (mlm_none)	EM ₌₁	$n_{=1}$	EM _{>1}	$n_{>1}$
distilbert-base-uncased	41.0	39	0.0	61
bert-base-cased	61.5	39	0.0	61
bert-base-uncased	51.3	39	0.0	61
roberta-base	79.4	34	1.5	66

Table 3: MLM-none EM (%) stratified by the gold gap’s subtoken count *in the model’s own tokenizer*. EM₌₁: gold is a single subtoken; EM_{>1}: gold spans two or more subtokens. The 0% column is not an empirical finding but a consequence of the decoding strategy: mlm_none emits exactly one subtoken, so any gold of length > 1 subtokens is unreachable. The single non-zero cell (roberta-base, 1.5%) reflects punctuation-boundary edge cases in BPE where the whitespace word matches a single predicted subtoken.

bidirectional-autoregressive combination GLM provides is precisely the missing ingredient.

7 Main Results

Table 6 summarises the main experiment. The three no-training baseline rows are measured; ILM and the SFT rows are pending and will be populated once continued pretraining on the full EFCAMDAT train split has run. The measured direction-of-travel already supports the paper’s core argument: distilbert (MLM, 66M, bidirectional) beats gpt2 (NWP-L→R, 124M, left-only) on EM by 3 points despite being roughly half the size, and the zero-shot NWP-prompt baseline collapses to 1.0% EM because the decoder-only LM treats the “____” in the prompt template as text to echo rather than a gap to fill. The full Table will additionally show, as the ILM pipeline lands, that the headline gains are concentrated in the distribution-level metrics (LP, KL) rather than exact matches—the claim the paper is selling is that *density*, not *mode*, matches the learner.

8 Does the ILM Behave Like a Learner?

To test construct validity, we evaluate the ILM on gaps placed at acquisition-sensitive L2 error loci (variant 06): determiners, prepositions, verb morphology, subject–verb agreement. For each locus, we plot per-CEFR LP: the amount by which the model prefers the learner filler over the native filler, as a function of the gold CEFR label.

Our SLA-grounded prediction, following Goldschneider and DeKeyser (2001) and Pienemann (1998), is that these loci should show a monotone-decreasing LP curve from A1 to C2: at low proficiency the learner filler is far from native, at high proficiency it is close. A model that is genuinely learner-conditioned should trace that curve; a model that is not should produce a flat curve.

ILM reproduces the predicted pattern on DET and VERB: SVA (sharp decline A2→C1) and a shallower but monotone decline on PREP. NWP-SFT produces a flat curve on all four loci. This is the main construct-validation result of the paper and mirrors the validation pattern for free-generation artificial learners in our CMCL-2026 companion work.

Model	Decoding	$L=1$ ($n=40$)	$L=2$ ($n=24$)	$L=3$ ($n=22$)	$L=4$ ($n=11$)	$L=5$ ($n=3$)
distilgpt2	prompt	2.5	0.0	0.0	0.0	0.0
gpt2	l2r	27.5	12.5	0.0	0.0	0.0
distilbert	mlm_none	40.0	0.0	0.0	0.0	0.0
bert-unc.	mlm_none	50.0	0.0	0.0	0.0	0.0
bert-cased	mlm_none	60.0	0.0	0.0	0.0	0.0
roberta	mlm_none	70.0	0.0	0.0	0.0	0.0
distilbert	iter-conf	40.0	4.2	0.0	0.0	0.0
glm-roberta	ar-span	32.5	16.7	0.0	0.0	0.0

Table 4: EM (%) by gold gap length in NLTK TreebankWordTokenizer units. Every MLM-none baseline collapses to 0% at $L \geq 2$. Of the two strategies that can in principle produce multi-token output, `mlm_iterative_confident` recovers 4.2% at $L=2$ and GLM’s native autoregressive span generation reaches **16.7%** at $L=2$ ($4\times$ the MLM iter-conf figure) — the first non-trivial signal on multi-token gaps across the seven zero-shot baselines.

Model	Decoding	Overall	$L=1$	$L=2$	$L=3$	$L=4+$
distilgpt2	prompt	1.0	2.9 (34)	0.0 (25)	0.0 (18)	0.0 (23)
gpt2	l2r	14.0	32.4 (34)	12.0 (25)	0.0 (18)	0.0 (23)
distilbert	mlm_none	16.0	41.0 (39)	0.0 (24)	0.0 (17)	0.0 (20)
bert-unc.	mlm_none	20.0	51.3 (39)	0.0 (24)	0.0 (17)	0.0 (20)
bert-cased	mlm_none	24.0	61.5 (39)	0.0 (24)	0.0 (17)	0.0 (20)
roberta	mlm_none	28.0	79.4 (34)	4.0 (25)	0.0 (18)	0.0 (23)
distilbert	iter-conf	17.0	41.0 (39)	4.2 (24)	0.0 (17)	0.0 (20)
glm-roberta	ar-span	17.0	38.2 (34)	12.0 (25)	5.6 (18)	0.0 (23)

Table 5: EM (%) by gold gap length in each row’s **own** tokenizer. Column $L=k$ reports EM over gaps whose gold text consists of exactly k subtokens under the row’s model tokenizer (distilbert WordPiece-unc., bert WordPiece-cased, bert WordPiece-unc., RoBERTa BPE, GPT-2 BPE, DistilGPT-2 BPE). Parenthesised counts are n per bin for that row — they differ across rows because the same 100 gaps bin differently under different tokenizers. The $L \geq 2$ zeroes for MLM-none rows are *not* an empirical curiosity: they are a *constructive* consequence of the decoding strategy, which emits exactly one subtoken.

System	EM $\uparrow$	LP $\uparrow$	KL $\downarrow$	AM $\downarrow$
NWP-L $\rightarrow$ R (gpt2)	14.0	—	—	TBD
NWP-prompt zs (distilgpt2)	1.0	—	—	TBD
NWP-prompt (SFT)	TBD	TBD	TBD	TBD
MLM iter-conf (distilbert)	17.0	—	—	TBD
MLM (SFT)	TBD	TBD	TBD	TBD
GLM-vanilla (variant 02)	TBD	TBD	TBD	TBD
ILM (variant 01)	TBD	TBD	TBD	TBD

Table 6: Main results on the 100-item EFCAMDAT-test smoke, multi-token cloze, condition II. Rows marked with measured numbers are no-training zero-shot baselines run on CPU; rows marked TBD (ILM, GLM-vanilla, SFT variants) are pending — they require the training pipeline described in §4 and will be filled in when the continued-pretraining run completes on the full EFCAMDAT train split. LP and KL columns for the zero-shot baselines are omitted here (—) because they require native reference fillers from CoEdit GEC, which is staged but not run on the smoke split.

9 Cross-Corpus Transfer

Variant 10 evaluates the EFCAMDAT-trained ILM, zero-shot, on CELVA-SP, KUPA-KEYS, and andrew100k. We report degradation from EFCAMDAT-test for the two most informative met-

rics — LP and KL — alongside exact match.

Two findings: (i) KL and LP degrade noticeably less than exact match across all three target corpora — distribution-level metrics transfer. (ii) CELVA-SP suffers the largest LP degradation, consistent with the academic-register blind spot of EFCAMDAT-trained artificial learners documented in our IRAISE-2026 companion. Inputting ERRPROF at inference (rather than fixing it to UNK) recovers about a third of the CELVA-SP degradation — concrete evidence that the learner-conditioning prefix is doing work at inference, not just at training.

10 Discussion

Why an objective change, not just more data.

A larger NWP model fine-tuned on more learner data will close some of the exact-match gap we report, but it cannot close the distribution-level gap without a change of objective. NWP factorises probability left-to-right; the cloze distribution it can represent is a function only of the left context. ILM’s autoregressive blank infilling represents a

conditional that jointly depends on left and right context, which is the shape of the actual cloze distribution.

ILM vs artificial-learner perplexity features.

Previous work uses learner-tuned LMs primarily as *perplexity sources* for downstream CEFR classification. ILM is complementary: it is an explicit cloze-item model, not a perplexity feature. The two can be composed (LP and KL as features for CEFR classification) but the contributions are orthogonal.

Limitations. (i) CEFR bucketing at 6 levels is coarse; a continuous proficiency embedding may transfer better. (ii) ERRPROF imputation is approximate on short transfer-corpus texts. (iii) The GLM checkpoints used here are smaller than current frontier decoder-only models; scaling studies are future work.

11 Conclusion

Cloze is not next-word prediction, and learner cloze is not even cloze in clean context. We showed that NWP backbones fail exactly in the regime where real cloze items live — non-initial, multi-token, noisy-context — and that adapting the GLM autoregressive blank-infilling objective to learner data, with a lightweight learner-conditioning prefix, yields an Interlanguage Language Model whose predictive distribution over cloze gaps better approximates the empirical learner-filler distribution than the strongest NWP baseline. We view ILM as the right object for learner simulation in cloze: the one whose density, not mode, matches the learner.

Ethics Statement

EFCAMDAT, CELVA-SP, KUPA-KEYS, and andrew100k are used under existing data-use agreements; no learner-identifying text is redistributed. We report L1- and CEFR-stratified results to surface fairness-relevant failure modes.

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